Colour and Memory: Colour as an aid to memorization

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Outline

- Goal of study
- Experiment
- Initial results
- Effect of colour and spatial attributes
- Conclusions

Colour and recall

Does the colour of an object affect the ability to recall it?

Yes

- Increases visual salience
- Enhances visual memory Dzulkifli et al, 2013, attention
- Enhances identification and recall of colour-coded information Carter et al, 1988
- Memory cue, strengthens neural encoding Spence et al, 2006

No

- Effects negligible without context Dzulkifli et al, 2013
- Positive results have weaknesses in experiment design
  - Semantically-related stimuli aiding recall
  - Insufficient control of time duration

Aim

- To investigate the degree to which colour enhances short-term object recall compared to grayscale stimuli
  - Using randomized, non-associated images to eliminate guessing bias
  - Using defined time periods for observation, retention and recall

Experiment

Based on ‘Kim’s Game’

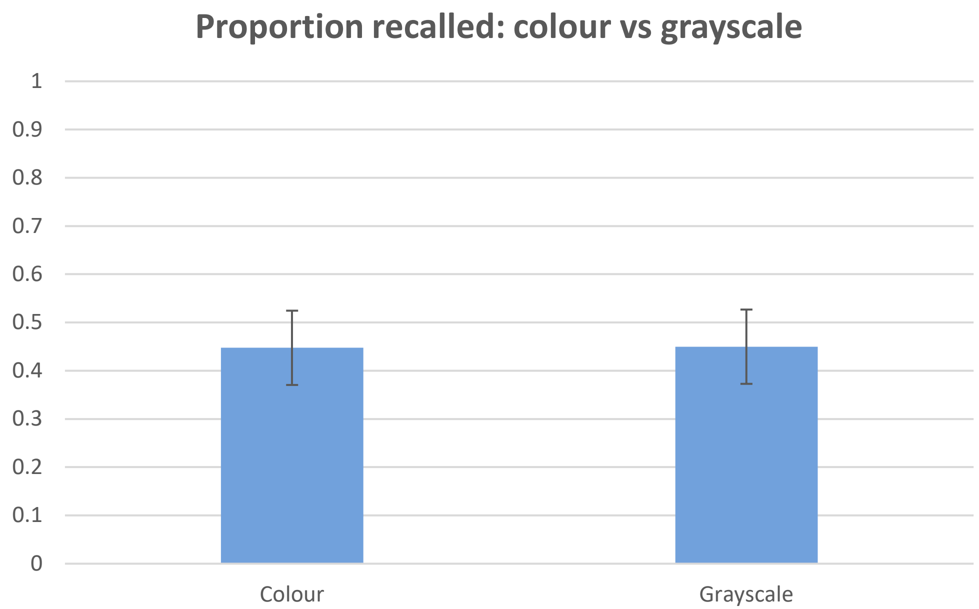
- **Encoding**: observers view an image containing multiple objects for 15 seconds
- **Retention**: a 10-second period with a distraction task
- **Recall**: observers list all remembered items, 60 second time limit

Two images: one grayscale and one colour

Stimuli

- Chat GPT prompt: "Generate a photorealistic image containing 20 clearly visible household objects arranged naturally on a table. Include items like a wristwatch, pen, and Post-it notes with varied colors and sizes, ensuring all objects are distinct and non-overlapping."

Results



Colour	Grayscale
0.448	0.450

Further analysis

- To understand the reason for these results, the effect of specific colour and spatial attributes was analyzed
  - L^* lightness
  - C^* chroma
  - h_{ab} hue angle
  - Size
  - Entropy
  - Contrast

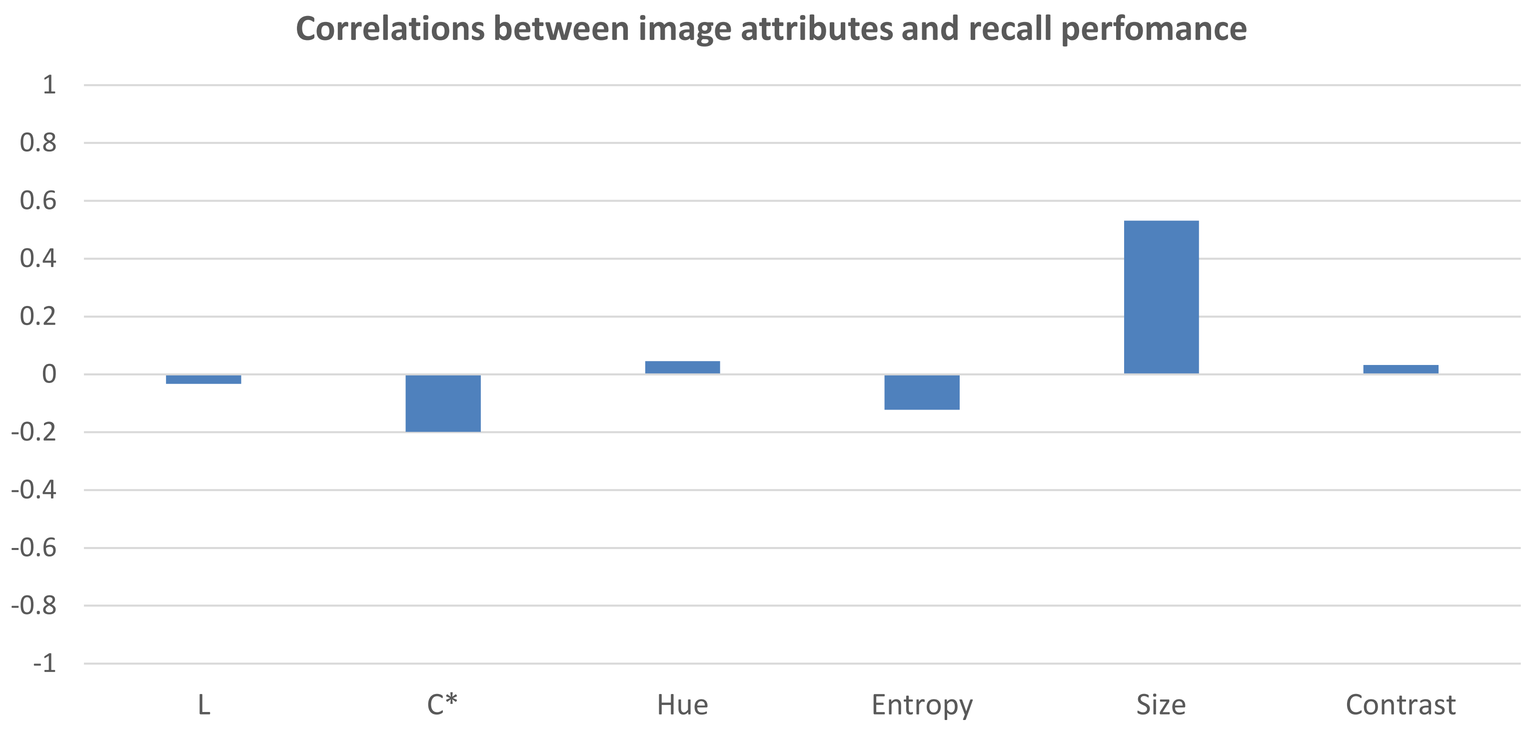
Image contrast

Modified Mitchelson contrast

$$c = (L_{max} - L_{min}) / (L_{max} + L_{min})$$

where L_{max} and L_{min} are replaced by 95^{th} and 5^{th} percentiles

Results



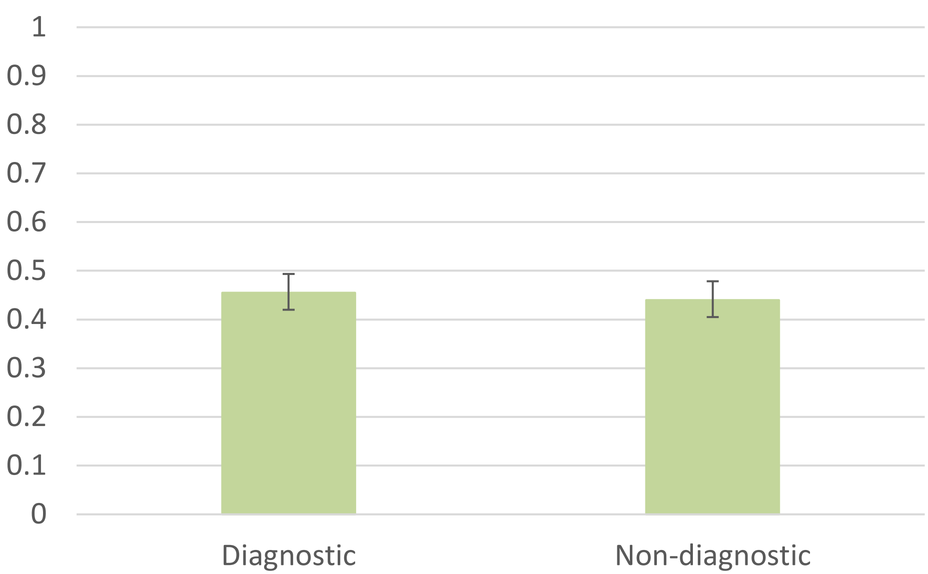
L^*	C_{ab}^*	h_{ab}	Entropy	Size	Contrast
-0.032	-0.199	0.046	-0.122	0.531	0.033

Further analysis

- Previous work has suggested that colour affects recall when it is strongly associated with an object and aids recognition
  - Diagnostic
  - Non-diagnostic

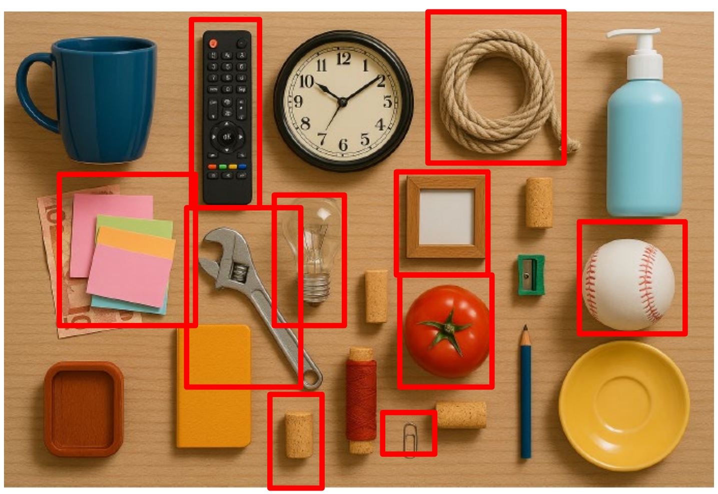
Results

Proportion recalled: effect of association



Diagnostic	Non-diagnostic
0.455	0.450

Objects where colour is associated and might be 'diagnostic' – i.e. aid in identification



Conclusions

- The effect of colour on short-term recall was investigated
- 20 participants were asked to remember 17-20 objects familiar household objects from an AI-generated image
- There was no significant different between colour and greyscale images
- Image attributes were analysed to determine their effect on recall
  - Colour attributes had no significant effect
  - Contrast and entropy had no significant effect
  - Size had the largest effect

Limitations

- Small study, only 20 participants
- Relatively small number of test objects and backgrounds
- Only specific short-term recall (10 seconds) tested
- Test items limited to ‘household objects’
- Image attributes not included in research hypothesis, used after the experiment to better understand results – could be studied more systematically

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Thank you!

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